AALBORG UNIVERSITY
Department of Computer Science

Software Engineering for AI Systems
Spring 2027

Portfolio 2: Cloud Continuum

Case: Short Anonymised Case Name

Student Name

October 27, 2026

Assessment criteria (delete this box before submitting):

- Placement decisions follow from the P1 quality attributes and constraints.
- Data flows respect the stated sovereignty rules, and the design enforces them.
- At least one security and one reliability tactic are demonstrated, not only described.
- A model can be rebuilt from Git, DVC and an MLflow run id, and one command deploys the system.

1 Deployment View

Show which component runs on which tier (edge, on-premises, cloud), and trace each placement to the P1 scenarios. Say how heterogeneous compute (small devices, CPUs, GPUs) constrains the choice of model.

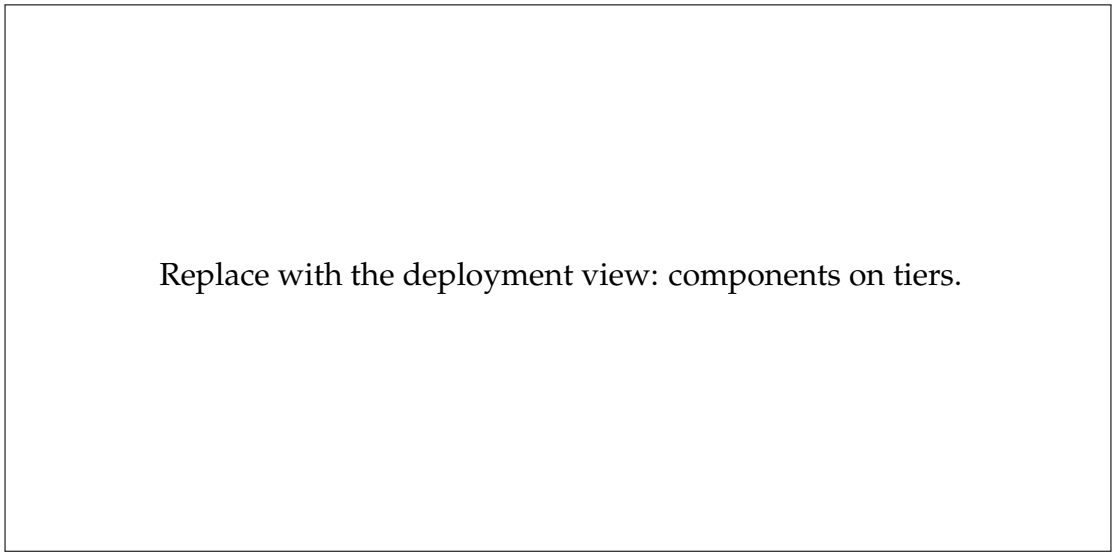


Figure 1: Deployment view.

Component	Tier	Why here	Scenario
…	…	…	QA-…
…	…	…	QA-…

Table 1: Component placement.

2 Data Placement

For each data item, state where it may be stored and processed, and the rule behind it (contract, GDPR, customer policy). Say how the design enforces the rule.

Data item	Stored at	Processed at	Rule
…	…	…	…
…	…	…	…

Table 2: Data placement.

3 Build or Buy

Compare at least one managed cloud or vendor ML platform with self-hosting [1], and record the decision as an ADR.

Criterion	Managed / vendor option	Self-hosted
Cost	…	…
Lock-in	…	…
Skills	…	…
Operations	…	…

Table 3: Managed versus self-hosted.

3.1 ADR-3: Title of the Decision

Status: Proposed / accepted.

Context: The problem and the scenarios (QA-…) that drive it.

Alternatives: The managed or vendor option, and self-hosting.

Decision: The chosen alternative.

Consequences: The trade-off that decided it, and what gets harder.

4 Security and Reliability Tactics

Map each tactic to the component it protects. Mark which tactics are demonstrated in Section 6. Say what happens when a part fails or the connection drops.

Tactic	Type	Component	Scenario	Demonstrated
…	Security	…	QA-…	Yes / no
…	Reliability	…	QA-…	Yes / no

Table 4: Security and reliability tactics.

5 Deployment and Versioning

Describe the Kubernetes manifests or Helm chart, the CI and deployment workflow, and how data and models are versioned. Show how a model is rebuilt from a Git commit, a DVC data version and an MLflow run id, and the one command that deploys the system.

6 Evidence

State whether the evidence comes from your own case or from the reference project. Give the repository (a private repository or a zip upload is fine), the output of `kubectl get all`, and a deleted pod or a failed model showing recovery.

```
$ kubectl get all
...
```

7 AI Usage

Briefly describe your use of AI tools on this assignment:

- *Code generation:* ...
- *Grammar and writing:* ...
- *Clarification of technical aspects:* ...

Bibliography

[1] L. Bass, Q. Lu, I. Weber, and L. Zhu, *Engineering AI systems*. Boston, MA: Addison Wesley, Mar. 2025.